Raphael Ramos

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Education

University of Alberta — Edmonton, AB
Bachelor of Science in Computer Engineering — *Sept. 2021 – June 2026*

Certifications

Engineer-in-Training (EIT), APEGA — 2026

Technical Skills

Languages: Python, C, C++, Java, HTML, VHDL, MATLAB, ARM, MIPS
Embedded & Hardware: STM32, STM32CubeIDE, Raspberry Pi Pico 2W, Xilinx Zybo Z7 FPGA, Edge Impulse, FreeRTOS, GDB, JTAG, Vivado, LTSpice, Cadence Virtuoso, WaveForms, KiCad, PCB Design, Fusion 360
Protocols: I2C, SPI, UART, PWM
Cloud & Backend: AWS IoT Core, Lambda, DynamoDB, Firebase, MQTT/TLS
Tools & Libraries: Git, GitHub, Docker, Android Studio, VS Code, Linux (CachyOS/Hyprland, Ubuntu, Fedora), NumPy, Pandas, Scikit-learn, OpenMP

Experience

Product Development Intern — May 2024 – Aug. 2024
TURBINE-X Energy Inc. — *Nisku, AB*
- Performed system-level testing and validation for fully integrated prototypes to ensure specification compliance.
- Calibrated advanced electrical, optical, and sensor equipment ensuring precise measurements across test campaigns.
- Built and brought up servers for internal data sharing, including physical construction and digital configuration.
- Created technical documentation for testing equipment covering training materials, specifications, and test conditions.

Software Deployment Intern — July 2023 – Sept. 2023
Vertical City — *Edmonton, AB*
- Updated Linux-based software from Ubuntu 18 to Ubuntu 20 across 250+ elevator and lobby screens.
- Troubleshot LTE signal processing with antennas, optimizing network connectivity for uninterrupted communication.
- Executed hardware re-wiring on elevator TVs and collaborated with supervisors and building technicians to coordinate deployment.

Projects

Continuing Care Home Activity Monitor | *C++, MicroPython, Edge Impulse, AWS IoT, React Native* — Jan. – Apr. 2026
- Built a non-invasive elder care monitoring system across 3 subsystems: stove detection, water usage, and appliance tracking.
- Deployed edge ML on a Raspberry Pi Pico 2W via Edge Impulse — 97.73% accuracy, sub-10-second response.
- Implemented full AWS IoT pipeline — MQTT/TLS, DynamoDB, Lambda anomaly detection, and Firebase push notifications.
- Delivered a caregiver React Native mobile app with real-time alerts and activity history; prototype under $200 CAD.

RP2040 Motor-Controller Carrier Board | *KiCad, PCB Design, Schematic Capture* — 2026
- Designing a custom 2-layer PCB to replace a breadboard prototype (Pico, H-bridge, pot, LED ring, encoder) — schematic capture and footprints in KiCad; layout, ground pour, and JLCPCB fab to follow.

Anomaly Detection System | *Python, Scikit-learn, Pandas, Matplotlib* — Apr. 2025
- Built a detection pipeline using statistical, distance-based (k-NN, DBSCAN), and ML (One-Class SVM) techniques.
- Applied preprocessing for missing value handling, normalization, and stratified train-test-validation splits.
- Evaluated models via precision, recall, F1-score, ROC/PR curves; documented end-to-end in a reproducible Jupyter notebook.

Event Lottery System | *Java, Firebase, Android Studio* — Sept. – Nov. 2024
- Built an Android app with a fair lottery-based sign-up system, Firebase Firestore backend, and local push notifications.
- Applied OOP design principles (CRC cards, UML) with comprehensive JUnit testing for scalability and maintainability.

16-bit CPU Design | *VHDL, Vivado, Xilinx Zybo Z7 FPGA* — Nov. 2024
- Designed a 16-bit CPU with controller-datapath architecture: ALU, register file, accumulators, and tri-state buffer.
- Developed an FSM executing 13 custom instructions; implemented a bitwise multiplier as a validation test program.
- Resolved double-access load discrepancies through rigorous debugging and simulation in Vivado.

Digital IC Design | *Cadence Virtuoso, CMOS, SPICE, WaveForms* — Fall 2024
- Designed and simulated a complete CMOS digital system in Cadence Virtuoso: parameterized inverter, NAND gates, 4-to-1 MUX, TSPC flip-flop, and 2-bit shift register.
- Integrated a thermometer-to-binary encoder with shift registers and MUX as a complete end-to-end digital system.
- Performed transient/DC analysis, transistor sizing, and process-corner/temperature and dynamic-power characterization.

Volunteer

Vice President, Academics — Apr. 2025 – Present
Computer Engineering Club, University of Alberta — *Edmonton, AB*
- Lead academic programming for 100+ members including workshops, study sessions, and exam preparation events.
- Coordinate with faculty and student groups to connect members with academic resources and opportunities.

Ground Station Software Team — Sept. 2023 – Sept. 2024
AlbertaSat, University of Alberta — *Edmonton, AB*
- Contributed to ground station software development for CubeSat operations and reliable data transmission.
- Collaborated with technical teams on implementation tasks and participated in regular development work sessions.